

Developing and Deploying Semantic Web Apps Using Open Source Tools

A semantic Web is a network in which data can be processed by machines. In other words, the data from the Web is machine readable because it is structured and tagged in such a way that machines can extract meaningful information in response to users' queries. Social media sites use semantic Web algorithms to make suggestions and create associations between persons.



Since the advent of online publishing and the digital marketing of products and services, search engines have been used to get accurate information. Nowadays, most services and products are searched for by people from search engines, using different search tricks and specific keywords.

According to data analytics reports of *InternetLiveStats.com*, Google processes more than 40,000 search queries per second. Around 20 per cent of these queries are new, having never been entered in the search engine before. The major challenge with search engines is to get accurate results without irrelevant outcomes.

The semantic Web broadly refers to 'a Web with meaning, using interconnections'. The 'Web' in a semantic Web is able to describe things in a way that computers can

understand the actual meaning that resides in the search query or browsing behaviour.

The wonderful powers of the semantic Web can be seen on websites like *Skyscanner.net* and *Trivago.com*. These portals compare the real-time prices from different service provider portals and give the best results in the form of price comparisons. The back-end libraries of these portals communicate with different websites and then fetch the results so that users can see the minimum price of flights or hotels. The protocols of the semantic Web work with these websites to fetch the related information from multiple locations.

In the semantic Web, the information on objects (music, vehicles, cars, persons, tickets, etc) is stored in special format files, and intelligent Web applications

that collect information from many different sources, analyse and present it to users in a meaningful way.

Whenever information about specific services, products, companies or objects is required, users navigate different search engines so that related websites can be fetched and the particular information can be found. One major issue with traditional search engines is that the information may be scattered and irrelevant.

For example, if a user enters the search keywords 'gold today' in the search bar, these may give many haphazard results including the following:

- Gold prices
- Star Gold
- Gold jewellers
- Gold loans

Similarly, the search word 'Python' may give results that include references